A Multimodal Benchmark for Layout-Aware Extraction

Synthetic document created for document-parsing evaluation.

Abstract

This synthetic study evaluates text, table, equation, and figure extraction under controlled layout variation. The source contains Greek symbols, subscripts represented as text, citations, and compact quantitative statements.

1. Method

Precision P and recall R are combined using the harmonic mean:

$$F1 = 2PR / (P + R)$$

A standardized measurement is calculated as z = (x - mu) / sigma, where mu is the sample mean and sigma is the sample standard deviation.

Dataset	Pages	Precision	Recall	F1
A	1,250	0.94	0.91	0.925
B	900	0.89	0.87	0.880
C	650	0.85	0.79	0.819

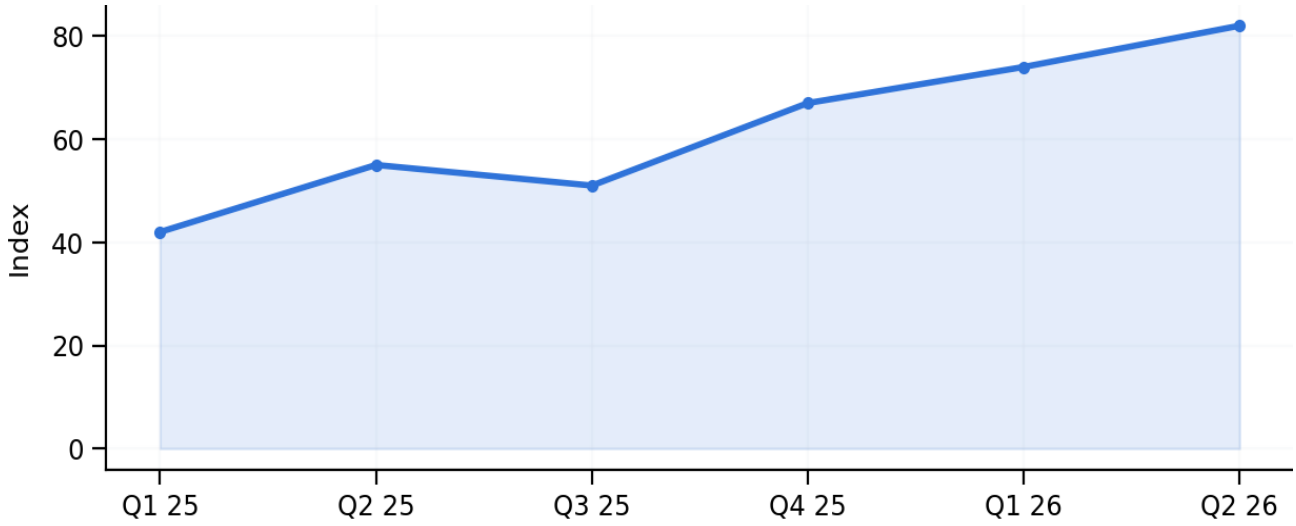

Figure 1. Synthetic quality index across six reporting quarters.

References

[1] Example Author. *Layout Evaluation Protocol*. 2025.
[2] Sample Research Group. *Structured Extraction Tests*. 2026.